

ARCHAEOLOGY



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the very beginning of the universe... Astronomers speak of “galactic archaeology” because they read the history of the cosmos from traces, just as archaeologists read ruins. They recognize the oldest stars in our galaxy because those stars contain very few elements heavier than helium, which formed only later. The cosmic background radiation is called “the echo of the Big Bang” – the oldest light we can see. Light needs time to travel, so a galaxy 10 billion light-years away appears as it was 10 billion years ago. Meteorites are “time capsules” because they contain the oldest matter: dust grains from the Murchison meteorite, older than the Sun, are about 7 billion years old. International teams share the James Webb Space Telescope, which shows the most distant galaxies as they were a few hundred million years after the Big Bang. Why do all this? To learn where the atoms in our bodies come from. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Astronomers call their study of the oldest light “galactic archaeology” because they...
 - a) read the history of the cosmos from traces
 - b) dig with shovels in space
 - c) study old telescopes
 - d) look for ruins on Mars
2. 👁 The oldest stars in our galaxy can be recognized because they contain very few...
 - a) light particles
 - b) hydrogen atoms
 - c) elements heavier than helium
 - d) neutrons
3. 🗨 The cosmic background radiation is sometimes called...
 - a) “the song of the Sun”
 - b) “the voice of the Earth”
 - c) “the roar of the ocean”
 - d) “the echo of the Big Bang”
4. ⏮ When we look at a galaxy 10 billion light-years away, we see it as it was...
 - a) a hundred years ago
 - b) tomorrow
 - c) 10 billion years ago
 - d) today
5. ✨ Scientists call meteorites “time capsules” because they...
 - a) contain the oldest matter
 - b) glow in the dark
 - c) are very hot
 - d) fall only in Poland
6. 👨🌍 International teams studying the oldest galaxies share the use of the...
 - a) James Webb Space Telescope
 - b) Copernicus Telescope
 - c) Newton Telescope
 - d) Galileo Telescope
7. 💖 Dust grains older than the Sun, found in the Murchison meteorite, are about...
 - a) 7,000 years old
 - b) 7 million years old
 - c) 7 billion years old
 - d) 70 years old
8. 🌿 The Webb telescope shows us the most distant galaxies as they were...
 - a) in the Middle Ages
 - b) just a few years ago, yesterday
 - c) before the dinosaurs
 - d) a few hundred million years after the Big Bang
9. ⭐ Studying the beginning of the universe helps answer the question...
 - a) who will win the match
 - b) what the weather will be tomorrow
 - c) where the atoms in our bodies come from
 - d) how much bread costs

PHILOSOPHY

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the moment when everything began, and listen to the philosophers. The Greek word “kosmos” originally meant order and ornament. The Pythagoreans believed in the “harmony of the spheres” – that the cosmos sounds like a music of numbers. Aristotle thought the universe had always existed and is eternal. Saint Augustine disagreed: time came into being together with the world, so asking what came “before” makes no sense. Gottfried Leibniz asked: “Why is there something rather than nothing?” Blaise Pascal confessed: “The eternal silence of these infinite spaces frightens me.” Immanuel Kant admired “the starry heavens above me and the moral law within me.” Georges Lemaître was both a physicist and a Catholic priest. Today the anthropic principle says the laws of physics are such that they allow observers to exist – that is, us. Remember these thoughts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The ancient Pythagoreans believed in the “harmony of the spheres” – the idea that the cosmos...
 - a) is made of water
 - b) was created yesterday
 - c) is silent and empty
 - d) sounds like a music of numbers
2. 👁 The question “Why is there something rather than nothing?” was asked by the philosopher...
 - a) Socrates
 - b) René Descartes
 - c) Gottfried Leibniz
 - d) Immanuel Kant
3. 🗨 The Greek word “kosmos” originally meant...
 - a) fire
 - b) darkness
 - c) order and ornament
 - d) water
4. ⚡ St. Augustine argued that time came into being...
 - a) never
 - b) together with the world
 - c) after humans
 - d) before the world
5. ⚡ Aristotle believed that the universe...
 - a) had always existed and is eternal
 - b) will vanish tomorrow
 - c) began with an explosion
 - d) is flat
6. 🧑 Georges Lemaître was both a physicist and a...
 - a) painter
 - b) soccer player
 - c) king
 - d) Catholic priest
7. 💖 Blaise Pascal wrote: “The eternal silence of these infinite spaces...”
 - a) delights me
 - b) frightens me
 - c) bores me
 - d) lulls me to sleep
8. 🌿 The anthropic principle says that the laws of physics are such that they...
 - a) change every day
 - b) are made up
 - c) do not work
 - d) allow observers to exist
9. ☆ Immanuel Kant wrote of two things that fill us with awe: “the starry heavens above me and...”
 - a) the city behind me
 - b) the sea beneath me
 - c) the garden in front of me
 - d) the moral law within me

HISTORY



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – and meet the people who discovered it. In 1929 Edwin Hubble showed that galaxies are moving away from us, so the universe is expanding. Earlier, Henrietta Swan Leavitt found a relationship that made it possible to measure distances to galaxies. The name “Big Bang” was coined in 1949 by Fred Hoyle, an opponent of the theory; with Bondi and Gold he promoted the steady-state theory – Hoyle, Bondi and Gold claimed the universe had no beginning. In 1965 Arno Penzias and Robert Wilson thought the noise in their antenna was caused by pigeons on the antenna, but they were hearing the background radiation. In 1992 the COBE satellite measured tiny temperature differences in it that were the seeds of galaxies. The first stars began to shine about 100–200 million years after the Big Bang. The universe is about 13.8 billion years old and Earth about 4.5 billion, so roughly 9 billion years passed before Earth formed. In 1951 Pope Pius XII called the theory consistent with the idea of creation. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In 1965, Arno Penzias and Robert Wilson thought the noise in their antenna was caused by...
 - a) pigeons on the antenna
 - b) a thunderstorm
 - c) a wrong musical note
 - d) a neighbor's radio
2. 👁 In 1929, Edwin Hubble showed that galaxies are...
 - a) disappearing
 - b) standing still
 - c) moving away from us
 - d) moving toward us
3. 📖 The name "Big Bang" was coined in 1949 by...
 - a) Fred Hoyle, an opponent of the theory
 - b) Isaac Newton
 - c) Edwin Hubble
 - d) Albert Einstein
4. ➡ The universe is about 13.8 billion years old and the Earth about 4.5 billion years. How many billion years passed before the Earth formed? Round to a whole number.

5. ✨ In 1992, the COBE satellite measured tiny temperature differences in the background radiation, which...
 - a) changed the color of the sky
 - b) revealed aliens
 - c) were the seeds of galaxies
 - d) were a measurement error
6. 🧑 The steady-state theory, a rival of the Big Bang, was championed by...
 - a) Hubble and Lemaître
 - b) Copernicus and Kepler
 - c) Hoyle, Bondi and Gold
 - d) Newton and Leibniz
7. ❤️ Henrietta Swan Leavitt discovered a relationship that made it possible to measure...
 - a) distances to galaxies
 - b) the age of rocks
 - c) the mass of the Earth
 - d) the temperature of the Sun
8. 🌿 The first stars began to shine about...
 - a) before the Big Bang
 - b) 100–200 million years after the Big Bang
 - c) 13 billion years after the Big Bang
 - d) yesterday
9. ☆ In 1951, Pope Pius XII declared the Big Bang theory to be...
 - a) forbidden
 - b) a fairy tale
 - c) consistent with the idea of creation
 - d) contrary to faith



LINGUISTICS

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and look at the words we use to describe it. The Polish word “wszechświat” is made of “all” and “world” – literally the whole world.

“Galaxy” comes from the Greek “gala,” meaning milk – hence the Milky Way. Despite its name, a light-year measures distance, not time. “Quasar” is short for the English term quasi-stellar object. The Polish names of the elements helium and hydrogen come from Greek words for “sun” and “water-forming”: helium from “helios,” the sun, and hydrogen because it forms water.

“Elements” themselves are substances that cannot be chemically broken down further. Dark matter is so called because it does not shine and cannot be seen directly. The official names of celestial bodies are set by the International Astronomical Union. And the oldest words about the beginning? The Bible opens: “In the beginning God created the heavens and the earth.” Remember these words – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The word “galaxy” comes from the Greek “gala”, meaning...
 - a) star
 - b) light
 - c) milk
 - d) road
2. 👁 The unit called a “light-year” measures...
 - a) mass
 - b) distance
 - c) brightness
 - d) time
3. 🗨 The Polish word for “universe” is literally a combination of two words meaning...
 - a) “wind” and “dawn”
 - b) “all” and “world”
 - c) “everyone” and “saints”
 - d) “candle” and “everything”
4. 📏 The term “quasar” is short for the English phrase...
 - a) quick clock
 - b) quasi-stellar object
 - c) square star
 - d) sour gas
5. ✨ The names of the elements helium and hydrogen come from...
 - a) Greek words for “sun” and “water-forming”
 - b) random choice
 - c) the names of Polish kings
 - d) the names of cities
6. 🌐 The International Astronomical Union sets...
 - a) the rules of soccer
 - b) official names of celestial bodies
 - c) the dates of holidays
 - d) telescope prices
7. ❤ Dark matter is called that because it...
 - a) is black like coal
 - b) does not shine and cannot be seen directly
 - c) is sad and gloomy
 - d) is visible only at night
8. 🌿 Chemical elements are substances that...
 - a) come first in the alphabet
 - b) glow in the dark
 - c) are always metals
 - d) cannot be chemically broken down further
9. ☆ The first words of the Bible are: “In the beginning God created...”
 - a) man
 - b) the sun and the moon
 - c) the heavens and the earth
 - d) the seas

LITERARY STUDIES



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – as poets and writers tell it. In Greek mythology, in the beginning there was Chaos. The Norse said the world was made from the body of the giant Ymir, and the Chinese said the giant Pangu separated the sky from the earth. A Hindu myth describes the world emerging from a golden egg. In Tolkien's "Silmarillion" the world is born from music (the Ainulindalë). Psalm 19 begins: "The heavens declare the glory of God." Dante ends the "Divine Comedy" with the love that moves the sun and the other stars. There are jokes too: in Lem's "Cyberiad," Trurl and Klapaucius build a machine that can make anything starting with the letter "n," and in Douglas Adams's "Hitchhiker's Guide to the Galaxy" the answer to the great question of life, the universe and everything is the number 42. Remember these stories – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In Tolkien's "The Silmarillion", the world is created from...
 - a) an egg
 - b) music (the Ainulindalë)
 - c) tears
 - d) an explosion
2. 👁 In Greek mythology, in the beginning there was...
 - a) Chaos
 - b) the Titan Atlas
 - c) Olympus
 - d) Zeus
3. 🗨 In Douglas Adams's "The Hitchhiker's Guide to the Galaxy", the answer to "the Ultimate Question of Life, the Universe, and Everything" is the number...
 - a) 13
 - b) 42
 - c) 7
 - d) 100
4. ⚡ In Norse mythology, the world was made from the body of the giant...
 - a) Odin
 - b) Loki
 - c) Ymir
 - d) Thor
5. 🛠 In Stanisław Lem's "The Cyberiad", the constructors Trurl and Klapaucius build, among other things, a machine that...
 - a) does not work at all
 - b) can make anything starting with the letter "n"
 - c) cooks soup for the whole kingdom
 - d) sings songs
6. 🐉 A Chinese myth says the world was created when the giant Pangu...
 - a) fell asleep
 - b) built a wall
 - c) separated the sky from the earth
 - d) ate the sun
7. ❤ Dante ends his journey in "The Divine Comedy" with words about the love that moves...
 - a) the seas
 - b) people
 - c) the sun and the other stars
 - d) the winds
8. 🌿 A Hindu myth describes the world coming out of...
 - a) a golden egg
 - b) a stone
 - c) a tree
 - d) water from ice
9. ☆ Psalm 19 begins with the words: "The heavens declare..."
 - a) the end of the world
 - b) the glory of God
 - c) the weather
 - d) the time

CULTURE AND RELIGION



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe as culture and religion have portrayed it. The biblical account of creation is found in the Book of Genesis. Michelangelo painted the fresco "The Creation of Adam" in the Sistine Chapel. In his 1798 oratorio "The Creation," Joseph Haydn depicted the words "And there was light" with a sudden, powerful chord. The rose windows of Chartres Cathedral and many churches symbolize an ordered cosmos. In his "Canticle of the Sun," Saint Francis calls the Sun his brother. The Vatican runs its own observatory, the Specola Vaticana. Other peoples have their own stories: for the Maya, creation is told in the book Popol Vuh, and Aboriginal Australians speak of the Dreamtime, when ancestors shaped the world by walking and singing. Terrence Malick's 2011 film "The Tree of Life" shows the birth of the universe alongside a family's story. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In his oratorio “The Creation” (1798), Joseph Haydn set the words “And there was light” to...
 - a) a sudden, powerful chord
 - b) a whisper
 - c) a whistle
 - d) silence
2. 👁 Michelangelo’s fresco “The Creation of Adam” is in the...
 - a) Cologne Cathedral
 - b) Louvre
 - c) Wawel Castle
 - d) Sistine Chapel
3. 📖 The biblical account of the creation of the world is found in the Book of...
 - a) Psalms
 - b) Revelation
 - c) Exodus
 - d) Genesis
4. 🏠 The Vatican runs its own astronomical observatory, called the...
 - a) Hubble
 - b) Arecibo
 - c) Mauna Kea
 - d) Specola Vaticana
5. 📖 In Maya culture, the creation of the world is described in the book...
 - a) Popol Vuh
 - b) the Edda
 - c) the Kalevala
 - d) Beowulf
6. 🧑 Aboriginal Australians tell of the Dreaming, when the ancestors...
 - a) flew rockets
 - b) built cities
 - c) fought robots
 - d) shaped the world by walking and singing
7. ❤ Terrence Malick’s film “The Tree of Life” (2011) shows, alongside a family story,...
 - a) a war in space
 - b) aliens
 - c) the birth of the universe
 - d) a sports competition
8. 🌿 Chartres Cathedral and many other churches have rose windows that symbolize...
 - a) a cartwheel
 - b) an eye
 - c) an ordered cosmos
 - d) the sun
9. ☆ In his “Canticle of the Sun”, St. Francis calls the Sun...
 - a) king
 - b) enemy
 - c) servant
 - d) brother

ECONOMICS AND FINANCE



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and count what it costs to study it. The James Webb Space Telescope cost about 10 billion dollars. Countries build such huge instruments together and share the costs and observing time. Telescope time is so precious that astronomers write proposals and compete for every hour. Poland has been a member of the European Space Agency since 2012, and it joined the European Southern Observatory in 2015. Why bother? Investment in basic science often leads to unexpected inventions, and supporters say that understanding the world matters to humanity. Space is also a treasury: gold formed, among other ways, in collisions of neutron stars, and the silicon and iron in our phones were made in stars long ago. Remember these numbers – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **The James Webb Space Telescope cost about...**
 - a) 100 dollars
 - b) 10 thousand dollars
 - c) 10 million zlotys
 - d) 10 billion dollars
2. 👁 **The most precious metals, such as gold, were formed in space partly during...**
 - a) collisions of neutron stars
 - b) meteor showers
 - c) sunrises
 - d) ordinary storms
3. 🗨 **Investment in basic science, such as space research, is justified by the fact that it...**
 - a) is cheap
 - b) often leads to unexpected inventions
 - c) brings immediate profit
 - d) gives nothing
4. 🇵🇱 **Poland has been a member of the European Space Agency (ESA) since...**
 - a) 2012
 - b) 2022
 - c) 1992
 - d) 1962
5. 🌌 **To build a large telescope, countries...**
 - a) fight each other
 - b) share the costs and observing time
 - c) draw lots for it
 - d) buy it in a store
6. 🌍 **Poland joined the European Southern Observatory (ESO) in...**
 - a) 2015
 - b) 1995
 - c) 1965
 - d) 2025
7. ❤ **Time on large telescopes is so valuable that astronomers...**
 - a) buy tickets
 - b) write proposals and compete for every hour
 - c) draw lots by phone
 - d) wait in line at the entrance
8. 🌿 **The elements that phones are made of, such as silicon and iron, were formed...**
 - a) in factories
 - b) last year
 - c) in Poland
 - d) in stars long ago
9. ⭐ **Is it worth spending billions on studying the beginning of the universe? Supporters say that...**
 - a) it achieves nothing
 - b) it is just a game
 - c) understanding the world matters to humanity
 - d) science should be banned

1.d 2.a 3.b 4.a 5.b 6.a 7.b 8.d 9.c

ANSWERS

PSYCHOLOGY



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and look inside the human mind. Children ask “Where did I come from?” very early. The vastness of space raises questions about the meaning of life for many people, and looking at space photos can stir wonder and humility (awe). The brain struggles to picture billions of years, so scientists use models such as Carl Sagan's “cosmic calendar”: if the Big Bang is January 1, humans appear on the evening of December 31. Data from space is also turned into sound so we can “listen” to the universe. We see animals and heroes in the stars because the brain likes to look for patterns. Watching an eclipse together makes people feel connected to others. And the “overview effect” is described by astronauts who have seen the Earth from space. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Data from space is sometimes turned into sound (sonification) so that we can...**
 - a) change the results
 - b) drown out scientists
 - c) "listen" to the universe
 - d) make an advertisement
2. 👁 **Looking at photos of space can evoke a feeling of...**
 - a) hunger
 - b) constant boredom
 - c) anger
 - d) wonder and humility (awe)
3. 🗨 **The "overview effect" is described by astronauts who have...**
 - a) seen the Earth from space
 - b) lost a match
 - c) fallen off a ladder
 - d) lost their keys
4. 🧠 **The human brain struggles to imagine billions of years, so scientists use...**
 - a) dreams
 - b) models such as the "cosmic calendar"
 - c) coloring books
 - d) fortune-telling
5. 🗨 **In Sagan's "cosmic calendar", where the Big Bang is January 1, humans appear on...**
 - a) the evening of December 31
 - b) Easter
 - c) February 1
 - d) a day in July
6. 👥 **Watching the sky together, for example during an eclipse, makes people...**
 - a) fall asleep
 - b) feel connected to others
 - c) argue
 - d) lose their memory
7. 💖 **Children usually ask the question "Where did I come from?"...**
 - a) only at university
 - b) only at school
 - c) never
 - d) very early
8. 🌿 **People see the shapes of animals and heroes in the stars because the brain...**
 - a) is too small
 - b) likes to look for patterns
 - c) counts stars
 - d) cannot see at night
9. ⭐ **For many people, realizing how vast the universe is...**
 - a) raises questions about the meaning of life
 - b) raises no questions at all
 - c) changes their eye color
 - d) makes them hungry



EDUCATION

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – and to how we learn about it. A planetarium is a place where the sky is projected onto a dome; the Copernicus Science Centre in Warsaw has the Heavens of Copernicus planetarium. Teachers explain that the Big Bang was not an explosion in space – space itself was expanding. A balloon with dots helps: as you inflate it, all the dots move away from each other. To show the scale of the Solar System, students can lay out a model of the planets on a sports field. A school astronomy club lets students observe the sky together, and Poland's Astronomy Olympiad has been held since 1957. Stephen Hawking wrote "A Brief History of Time" so that any curious reader could understand it. Many scientists say the most important thing in science is asking questions. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎫 **A planetarium is a place where...**
 - a) an orchestra plays
 - b) plants are grown
 - c) rockets are repaired
 - d) the sky is projected onto a dome
2. 👁 **The Copernicus Science Centre in Warsaw has...**
 - a) a zoo
 - b) an Olympic swimming pool
 - c) an airport
 - d) the Heavens of Copernicus planetarium
3. 🗨 **Teachers often explain the name “Big Bang” by stressing that...**
 - a) the center of the Earth exploded
 - b) space itself was expanding
 - c) it could be heard in space
 - d) a bomb exploded
4. 🎈 **A balloon with dots explains the expansion of the universe because, as it is inflated,...**
 - a) all the dots move away from each other
 - b) the dots disappear
 - c) the balloon always bursts
 - d) the dots move closer together
5. 📏 **To show the scale of the Solar System, students can...**
 - a) fit the whole model into a box
 - b) draw it on a fingernail
 - c) lay out a model of the planets on a sports field
 - d) eat it
6. 🌱 **A school astronomy club allows students to...**
 - a) grow plants
 - b) cook
 - c) observe the sky together
 - d) play soccer
7. 💖 **Cosmologist Stephen Hawking wrote “A Brief History of Time” so that it could be understood by...**
 - a) no one
 - b) only professors
 - c) any curious reader
 - d) only computers
8. 🍀 **The Astronomy Olympiad in Poland has been held since...**
 - a) 2017
 - b) 1957
 - c) 1857
 - d) 1997
9. ⭐ **Many scientists say that the most important thing in science is...**
 - a) never making mistakes
 - b) knowing all the answers
 - c) memorizing dates
 - d) asking questions

SOCIOLOGY

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and see how the sky brings people together. Creation myths helped communities feel who they are and where they come from. Today various religions and science answer the question of how the world began – it matters that we respect different views and talk. Big projects like the Webb telescope involve thousands of people from many countries. The Zooniverse project lets anyone help scientists classify galaxies, and the community of amateur astronomers watches the sky constantly. Sadly, light pollution in cities means most stars cannot be seen; people in the countryside, far from lights, can see the Milky Way. Poland's first dark-sky park was the Iżera Dark Sky Park. And the Night of Shooting Stars, the peak of the Perseids, falls in August. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Zooniverse project allows anyone to...**
 - a) play music
 - b) order food
 - c) help scientists classify galaxies
 - d) buy stars
2. 👁 **Light pollution in cities means that...**
 - a) there are more stars
 - b) the Moon disappears
 - c) the night is darker
 - d) most stars cannot be seen
3. 🗨 **Poland's first dark-sky park was...**
 - a) Royal Łazienki Park
 - b) Tatra National Park
 - c) the Iżera Dark Sky Park
 - d) the Saxon Garden
4. 🧑🔬 **The community of amateur astronomers is important because it...**
 - a) watches the sky constantly
 - b) buys telescopes
 - c) bans research
 - d) has no importance
5. ✨ **The Night of Shooting Stars, the peak of the Perseids, falls in...**
 - a) January
 - b) November
 - c) August
 - d) April
6. 👥 **Large projects such as the Webb telescope involve...**
 - a) two people
 - b) only robots
 - c) only one country
 - d) thousands of people from many countries
7. 💖 **Creation myths helped communities to...**
 - a) count their money
 - b) feel who they are and where they come from
 - c) raise animals
 - d) build roads and bridges
8. 🌿 **People living in the countryside, far from lights, can see...**
 - a) only the Moon
 - b) nothing
 - c) fewer stars than in the city
 - d) the Milky Way
9. ⭐ **Different religions and science answer the question of how the world began. It is important to...**
 - a) stop thinking
 - b) forbid questions
 - c) mock others
 - d) respect different views and talk

LAW



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and see what laws apply to the sky. The laws of physics, like the law of universal gravitation, differ from the laws of states because they apply everywhere and cannot be broken. Human law reaches space too. The 1967 Outer Space Treaty states that space is the common heritage of humankind, and placing nuclear weapons in orbit is forbidden. Cooperation on the International Space Station is governed by an agreement among the governments of many countries. Can you “buy a star”? No – it has no legal force. The night sky is protected by rules limiting light pollution, and a 2007 UNESCO declaration says the sky is the heritage of humankind. The Morasko meteorite craters near Poznań are protected as a nature reserve. And scientists have a duty of honesty: they do not falsify research results. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚫 **Bans on loud music at night protect silence, and the night sky is protected by rules on...**
 - a) limiting light pollution
 - b) store opening hours
 - c) banning telescopes
 - d) electricity prices
2. 🌌 **The Outer Space Treaty (1967) states that space...**
 - a) is closed
 - b) can be sold
 - c) belongs to the USA
 - d) is the common heritage of humankind
3. 📜 **Laws of physics, such as the law of universal gravitation, differ from laws of a country in that they...**
 - a) can be changed by referendum
 - b) apply only to people
 - c) apply everywhere and cannot be broken
 - d) are passed by parliament
4. ⚖️ **Under the Outer Space Treaty, placing nuclear weapons in orbit is...**
 - a) required
 - b) allowed
 - c) forbidden
 - d) not mentioned
5. ⭐ **Can you “buy a star” and give it your name? The answer is...**
 - a) yes, in any store
 - b) no – it has no legal force
 - c) only in Poland
 - d) yes, at a government office
6. 🤝 **Cooperation on the International Space Station is governed by...**
 - a) the law of a single small city
 - b) an agreement among the governments of many countries
 - c) the rules of one school
 - d) the highway code
7. ❤️ **Scientists have a duty of honesty, which means they...**
 - a) are always right
 - b) never make mistakes
 - c) do not falsify research results
 - d) hide data
8. 🌿 **The craters left by the Morasko meteorite near Poznań (about 5,000 years ago) are now protected as a...**
 - a) airport
 - b) parking lot
 - c) mine
 - d) nature reserve
9. ⭐ **The 2007 UNESCO declaration on the right to a dark sky says that the sky...**
 - a) is unnecessary
 - b) is forbidden
 - c) belongs to airports
 - d) is the heritage of humankind

OTHER (HUMANITIES)



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and see how it has enchanted people. Richard Strauss's "Also sprach Zarathustra" became famous through the film "2001: A Space Odyssey." Carl Sagan's 1980 series "Cosmos" opened with: "The Cosmos is all that is, or ever was, or ever will be." Sagan also said: "We are a way for the universe to know itself." The phrase "We are made of star stuff" means the elements in us were formed in stars – carbon, the basis of life, was made inside stars. The photo "Pillars of Creation" was taken by the Hubble telescope. In "Star Wars," hyperspace lets ships fly faster than light, though real physics forbids it. The universe is about 13.8 billion years old and the Sun about 4.6 billion, so the universe is about 3 times older. And science centers help families discover the laws of nature together. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Richard Strauss's "Also sprach Zarathustra" is famous from the film...**
 - a) "2001: A Space Odyssey"
 - b) "The Lion King"
 - c) "Titanic"
 - d) "Shrek"
2. 📷 **The "Pillars of Creation" photo, showing columns of gas where stars are born, was taken by the...**
 - a) Arecibo telescope
 - b) Chorzów telescope
 - c) Hubble telescope
 - d) Galileo telescope
3. 🗣️ **Carl Sagan's series "Cosmos" (1980) opened with the words: "The Cosmos is all that is, or was, or..."**
 - a) can be bought
 - b) ever will hurt
 - c) can be seen
 - d) ever will be
4. 📏 **The universe is about 13.8 billion years old and the Sun about 4.6 billion years. About how many times older is the universe than the Sun (whole number)?**

5. ✨ **In "Star Wars", travel through hyperspace lets ships...**
 - a) fly faster than light
 - b) go back in time
 - c) vanish forever
 - d) change shape
6. 🧑‍🔬 **Science centers and museums such as the Eksperymentarium help families...**
 - a) discover the laws of nature together
 - b) go shopping
 - c) play cards
 - d) look only at paintings
7. 💖 **Carl Sagan said: "We are a way for the universe to..."**
 - a) come to an end
 - b) take a rest
 - c) know itself
 - d) have fun
8. 🌿 **Carbon, the basis of life, was formed...**
 - a) inside stars
 - b) in the first second
 - c) on Earth
 - d) in ice
9. ⭐ **The famous phrase "We are made of star stuff" means that...**
 - a) we came from Mars
 - b) we are made of sand
 - c) the elements in us were formed in stars
 - d) we glow at night



ARCHITECTURE AND URBAN PLANNING

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – or rather to the buildings that let us see it. Observatories are built on high mountains because the air there is cleaner and drier. Chile's Atacama Desert is one of the driest places on Earth; it hosts ALMA, a network of 66 radio antennas, and the Extremely Large Telescope is being built with a mirror about 39 m across. An observatory dome rotates and opens so the telescope can point in different directions. China's FAST radio telescope from 2016 has a dish 500 m across. The observatory in Toruń (Piwnice) is famous for Poland's largest radio telescope. Neutrino detectors such as IceCube are placed deep in the Antarctic ice. And planned observatories on the far side of the Moon would enjoy radio silence from Earth's signals. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The FAST radio telescope in China (2016) has a dish with a diameter of...**
 - a) 50 km
 - b) 5 m
 - c) 5 cm
 - d) 500 m
2. 👁 **Astronomical observatories are often built on high mountains because...**
 - a) the air there is cleaner and drier
 - b) they are half as far from the stars
 - c) it is noisier there
 - d) it is warmer there
3. 🗨 **The dome of an astronomical observatory rotates and opens so that...**
 - a) rain can get in
 - b) music can be played
 - c) the telescope can point in different directions
 - d) people inside can cool off
4. 📏 **The Extremely Large Telescope (ELT) being built in Chile is to have a mirror with a diameter of about...**
 - a) 3 m
 - b) 3 cm
 - c) 39 m
 - d) 390 m
5. ❄ **Neutrino detectors such as IceCube are placed...**
 - a) on the roofs of houses
 - b) only in space
 - c) on beaches
 - d) deep in the Antarctic ice
6. 📡 **The ALMA observatory in Chile is...**
 - a) a planetarium
 - b) a TV tower
 - c) a network of 66 radio antennas
 - d) a single small telescope
7. 💖 **The Astronomical Observatory in Toruń (Piwnice) is famous for...**
 - a) its cinema
 - b) Poland's largest radio telescope
 - c) its zoo
 - d) its swimming pool
8. 🌿 **The observatories in Chile's Atacama Desert are special because...**
 - a) there is a lake there
 - b) there is a lot of rain there
 - c) there is a forest there
 - d) it is one of the driest places on Earth
9. ☆ **Planned observatories on the far side of the Moon would have...**
 - a) cheaper electricity
 - b) a view of the Earth
 - c) radio silence from Earth's signals
 - d) more rain

1.d 2.a 3.c 4.c 5.d 6.c 7.b 8.d 9.c

A N S W E R S



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and hear how electronics “listens” to it. Radio telescopes pick up radio waves from space, which can be turned into sound and images. Linked in a network like the Event Horizon Telescope, they act as a single Earth-sized telescope. CCD sensors turn light into an electrical signal. The Webb telescope observes in infrared because light from the farthest galaxies has been stretched along the way, and its detectors are cooled below $-260\text{ }^{\circ}\text{C}$. The solar panels of space telescopes are turned toward the Sun. The Planck satellite measured the background radiation to millionths of a degree. Cosmic radiation can damage satellite electronics, so shielding and hardened circuits are used. And part of the static on old analog TVs came from the echo of the Big Bang. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Radio telescopes receive radio waves from space, which can be turned into...**
 - a) smells
 - b) sound and images
 - c) home heating
 - d) tastes
2. 👁 **CCD sensors in telescopes, similar to those in cameras, convert...**
 - a) electricity into heat
 - b) water into ice
 - c) sound into an image
 - d) light into an electrical signal
3. 🗨 **The detectors in the Webb telescope are cooled to a temperature of...**
 - a) below $-260\text{ }^{\circ}\text{C}$
 - b) $+100\text{ }^{\circ}\text{C}$
 - c) $0\text{ }^{\circ}\text{C}$
 - d) $+20\text{ }^{\circ}\text{C}$
4. 📏 **The Webb telescope observes in infrared because the light from the most distant galaxies...**
 - a) is too bright
 - b) has been stretched along the way
 - c) is green
 - d) does not exist
5. 🌞 **The solar panels of space telescopes are turned toward...**
 - a) Mars
 - b) the Earth
 - c) the Sun
 - d) the Moon
6. 🧑 **Many radio telescopes linked in a network (e.g., the Event Horizon Telescope) work like...**
 - a) a calculator
 - b) a single Earth-sized telescope
 - c) a small pair of binoculars
 - d) a car radio
7. 💖 **The Planck satellite measured the background radiation with a precision of...**
 - a) millionths of a degree
 - b) hundreds of degrees
 - c) whole degrees
 - d) tens of degrees
8. 🍃 **Cosmic radiation can damage satellite electronics, so...**
 - a) they are painted black
 - b) satellites are made of paper
 - c) shielding and hardened circuits are used
 - d) they are switched off at night
9. ☆ **Part of the static on old analog TV sets came from...**
 - a) the neighbors
 - b) the rain
 - c) the echo of the Big Bang
 - d) the radio

COMPUTER SCIENCE AND TELECOMMUNICATIONS

**1. Beginning of the Universe**

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and see how computing helps us understand it. Computer simulations such as “Millennium” show how tiny differences in density gave rise to the cosmic web of galaxies. Artificial intelligence helps astronomers find objects in huge data sets. The 2019 black hole image needed so much data that hard drives were shipped by plane, and computer scientist Katherine Bouman helped create an algorithm for imaging a black hole. Communication is a challenge: Voyager 1 transmits at about 20 W – like a refrigerator light bulb – and a signal from Earth takes almost a full day to reach it. We send messages too: Pioneer 10 carried a plaque with a drawing of humans and Earth's location, and the 1974 Arecibo message contained numbers, DNA and a human figure. The SETI program searches for signals from alien civilizations. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **The Voyager 1 probe sends signals to Earth with a power of about 20 W – like...**
 - a) a power plant
 - b) an airplane
 - c) a refrigerator light bulb
 - d) a lightning bolt
2. 👁 **Computer simulations such as “Millennium” show how tiny differences in density gave rise to...**
 - a) the cosmic web of galaxies
 - b) water
 - c) the Moon
 - d) the Earth
3. 📠 **The 2019 image of a black hole was made from so much data that...**
 - a) it fit into a text message
 - b) it was sent by carrier pigeon
 - c) it was printed out
 - d) hard drives were shipped by plane
4. 📡 **A radio signal sent from Earth reaches Voyager 1 after about...**
 - a) one second
 - b) almost a full day
 - c) one minute
 - d) one year
5. 🌟 **Katherine Bouman, a computer scientist, helped create...**
 - a) an algorithm for imaging a black hole
 - b) Wikipedia
 - c) a telephone
 - d) a computer game
6. 🎧 **The SETI program searches for...**
 - a) new songs
 - b) signals from alien civilizations
 - c) lost ships
 - d) oil deposits
7. 💖 **Artificial intelligence helps astronomers...**
 - a) write poems about stars
 - b) find objects in huge data sets
 - c) sweep the observatory
 - d) point telescopes by hand
8. 🍃 **The Pioneer 10 probe (1972) carried a plaque with...**
 - a) a shopping list
 - b) a drawing of humans and Earth’s location
 - c) sheet music
 - d) a map of Warsaw
9. 🌟 **The Arecibo message (1974), sent into space, contained among other things...**
 - a) an advertisement
 - b) a photo of a cat
 - c) the US national anthem
 - d) numbers, DNA and a human figure

BIOMEDICAL ENGINEERING



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – antimatter already appeared in its first moments. Today PET scans use positrons, particles of antimatter, and detectors such as PET help detect cancer. Polish scientists in Kraków are developing the J-PET scanner made of cheap plastic scintillators. Accelerators, built to study the universe, now treat cancer in hospitals; proton therapy uses a proton beam from an accelerator. Iodine, used in medicine, was made in exploding stars. Engineers also look after astronauts. In weightlessness muscles and bones weaken, so astronauts exercise every day. Cosmic radiation can damage the DNA in cells. And long missions require psychological support, because astronauts live in isolation, far from loved ones. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Astronauts in orbit exercise every day because in weightlessness...**
 - a) muscles and bones weaken
 - b) they cannot hear
 - c) they grow too fast
 - d) they cannot sleep
2. 👁 **Cosmic radiation is dangerous for astronauts because it can...**
 - a) cure diseases
 - b) change hair color
 - c) give strength
 - d) damage the DNA in cells
3. 🗨 **Particle detectors from physics, such as PET scanners, help doctors...**
 - a) cook
 - b) cure a runny nose
 - c) detect cancer
 - d) measure height
4. ⚖ **PET scanning uses positrons, which are particles of...**
 - a) sound
 - b) water
 - c) light
 - d) antimatter
5. ✨ **Polish scientists in Kraków are developing the J-PET scanner made of...**
 - a) gold
 - b) wood
 - c) cheap plastic scintillators
 - d) window glass
6. 🧑 **Proton therapy, used in Kraków among other places, treats cancer with...**
 - a) sunlight
 - b) magnets
 - c) a proton beam from an accelerator
 - d) sound
7. 💖 **Astronauts on long missions need psychological support because they...**
 - a) dislike each other
 - b) get bored
 - c) live in isolation, far from loved ones
 - d) have no phones
8. 🌿 **Iodine, an element used in medicine, was formed in...**
 - a) a factory
 - b) a laboratory
 - c) the ocean
 - d) exploding stars
9. ⭐ **Particle accelerators, built to study the universe, are now used in hospitals to...**
 - a) play music
 - b) treat cancer
 - c) replace doctors
 - d) produce food

CHEMICAL ENGINEERING



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe, into its chemistry lab. After the Big Bang, mainly two elements formed: hydrogen and helium. Elements heavier than iron mostly form during supernova explosions. The oxygen we breathe was also made in stars. Cosmic dust contains grains of silicates and carbon, and organic molecules have been found in space, such as alcohol and simple sugars. We learn what stars are made of through spectroscopy – from the lines in their spectrum. Nebulae glow in various colors because different elements glow in different colors. Sound does not travel in space, because a vacuum has no molecules to vibrate. And antimatter is a puzzle: there is very little of it today, although there should be as much of it as matter. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Sound does not travel in space because...**
 - a) a vacuum has no molecules to vibrate
 - b) it is too cold
 - c) stars absorb it
 - d) it is too dark
2. 👁 **Nebulae glow in different colors because different...**
 - a) phones light them up
 - b) stars paint them
 - c) elements glow in different colors
 - d) clouds reflect them
3. 🗨 **Spectroscopy reveals what stars are made of, based on...**
 - a) the lines in their spectrum
 - b) their smell
 - c) their weight
 - d) their sound
4. ⚖ **After the Big Bang, two elements were formed in the greatest amounts:**
 - a) nitrogen and sodium
 - b) hydrogen and helium
 - c) oxygen and carbon
 - d) iron and gold
5. 🍫 **Organic molecules have been discovered in space, for example...**
 - a) chocolate
 - b) alcohol and simple sugars
 - c) bottled gasoline
 - d) plastic
6. 🌌 **Most elements heavier than iron are formed during...**
 - a) earthquakes
 - b) moonlight
 - c) rain
 - d) supernova explosions
7. 💖 **The oxygen we breathe was formed in...**
 - a) forests only
 - b) the sea
 - c) stars
 - d) a volcano
8. 🌿 **Cosmic dust contains grains of...**
 - a) table salt
 - b) sugar
 - c) silicates and carbon
 - d) flour
9. ⭐ **There is very little antimatter in today's universe. This is a mystery because...**
 - a) nobody has looked for it
 - b) there should be as much of it as matter
 - c) it is in every home
 - d) there is too much of it

CIVIL ENGINEERING AND TRANSPORT



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and set off with the Webb telescope. It launched on an Ariane 5 rocket from the European Space Agency's spaceport in French Guiana. Unfolding the telescope took about two weeks, and its sunshield is the size of a tennis court. Webb works at the L2 point, about 1.5 million km from Earth. Other probes set records too: Parker Solar Probe, the fastest human-made object, reaches about 190 km/s. Many probes use a gravity assist – they speed up by flying past planets. Solar sails are pushed by the pressure of sunlight. And the stars are very far away: light needs about 4.2 years to reach Proxima Centauri. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The Webb telescope was launched into space on the rocket...
 - a) Soyuz
 - b) Saturn V
 - c) Ariane 5
 - d) Falcon 9
2. 👁 The Webb telescope operates at the L2 point, which is about...
 - a) on the Moon
 - b) on Mars
 - c) 400 km from Earth
 - d) 1.5 million km from Earth
3. 🗨 The Webb telescope's sunshield is the size of...
 - a) an umbrella
 - b) a tennis court
 - c) a sheet of paper
 - d) a hundred basketball courts
4. ⚡ To reach the nearest star, Proxima Centauri, light needs about...
 - a) 4 days
 - b) 4 minutes
 - c) 400 years
 - d) 4.2 years
5. ⚡ Unfolding the Webb telescope after launch took about...
 - a) two weeks
 - b) two seconds
 - c) two years
 - d) two hours
6. 🧑 The European Space Agency's spaceport is located in...
 - a) Spain
 - b) Norway
 - c) French Guiana
 - d) Poland
7. ❤ The Parker Solar Probe is the fastest object ever built by humans – it reaches about...
 - a) 19 m/s
 - b) 190 km/s
 - c) 1900 km/s
 - d) 190 km/h
8. 🍃 Many probes use a “gravity assist”, which means they...
 - a) speed up by flying past planets
 - b) are pushed by the wind
 - c) use paper sails
 - d) are pulled by a rope
9. ☆ Solar sails are propelled by...
 - a) electricity from Earth
 - b) the pressure of sunlight
 - c) Earth's wind
 - d) gasoline

MATERIALS ENGINEERING



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and look at the materials that let us see it. The Webb telescope's main mirror consists of 18 hexagonal segments made of a light metal, beryllium, and coated with a thin layer of gold. To fit the telescope into the rocket, the mirror and shield were folded like origami. The shield has 5 layers because each layer cools it further. Lenses were once ground by hand; today telescope mirrors are polished with nanometer precision. Dark matter detectors are built from very pure materials because impurities would disturb the measurement. Iron meteorites consist mainly of iron and nickel. And aerogel on the Stardust probe caught comet dust because it is light and slows the particles down. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The mirrors of the Webb telescope are coated with a thin layer of...**
 - a) copper
 - b) silver
 - c) lead
 - d) gold
2. 👁 **Webb's primary mirror consists of...**
 - a) a single round mirror
 - b) 18 hexagonal segments
 - c) 100 squares
 - d) 2 semicircles
3. 🗨 **The Webb telescope's mirrors are made of a lightweight metal called...**
 - a) beryllium
 - b) lead
 - c) solid gold
 - d) iron
4. ➡ **Webb's sunshield has 5 layers because...**
 - a) each layer cools it further
 - b) it looks nice
 - c) it was cheap
 - d) it protects against rain
5. ⚡ **To fit the telescope into the rocket, its mirror and sunshield were...**
 - a) rolled up
 - b) inflated
 - c) cut into pieces
 - d) folded like origami
6. 👤 **The lenses of the first telescopes were ground by hand. Today, telescope mirrors are...**
 - a) made of plastic
 - b) bought in a store
 - c) painted with paint
 - d) polished with nanometer precision
7. ❤ **Dark matter detectors are built from extremely pure materials because...**
 - a) impurities would disturb the measurement
 - b) pure ones look nicer
 - c) they are cheaper
 - d) the law requires it
8. 🌿 **Iron meteorites consist mainly of...**
 - a) carbon
 - b) gold
 - c) iron and nickel
 - d) ice
9. ☆ **The aerogel on the Stardust probe captured comet dust because...**
 - a) it is light and slows the particles down
 - b) it is heavy
 - c) it is sticky
 - d) it is hot

1.d 2.b 3.a 4.a 5.d 6.d 7.a 8.c 9.a

A N S W E R S

MECHANICAL ENGINEERING



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and step into the workshop of space mechanics. Telescope mirrors are adjusted precisely with actuators that move the segments. Gyroscopes in the Hubble telescope are used for precise pointing. Hubble was repaired in space five times, with astronauts working in spacesuits, attached to a robotic arm. Atomic clocks on satellites keep time to billionths of a second. A Polish company built mechanisms and instruments for probes on ESA missions. In 2014 the Rosetta probe sent the Philae lander to a comet. Comets are like “freezers” because they contain ice and dust from when the planets formed. And Earth's rotation is shown by a Foucault pendulum – in Poland, for example, in Kraków's Church of Sts. Peter and Paul. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Telescope mirrors can be precisely adjusted using...**
 - a) hammers
 - b) ropes
 - c) actuators that move the segments
 - d) an astronaut's hand
2. 👁 **Atomic clocks on satellites measure time with a precision of...**
 - a) billionths of a second
 - b) a minute
 - c) a day
 - d) an hour
3. 📻 **The gyroscopes in the Hubble telescope are used for...**
 - a) recording sound
 - b) power supply
 - c) precise pointing
 - d) cooling
4. 🛠 **The Hubble telescope was repaired in space...**
 - a) once
 - b) never
 - c) a hundred times
 - d) five times
5. 🚀 **Astronauts repairing Hubble worked...**
 - a) underwater
 - b) in ordinary clothes
 - c) in an airplane
 - d) in spacesuits, attached to a robotic arm
6. 🧑 **Polish companies have built for ESA missions, among other things,...**
 - a) complete Ariane engines
 - b) mechanisms and instruments for probes
 - c) the Webb telescope
 - d) lunar rovers
7. 💖 **In 2014, the Rosetta probe sent the Philae lander to...**
 - a) a comet
 - b) the Moon
 - c) Mars
 - d) the Sun
8. 🌿 **Comets are like “freezers” from the early Solar System because they contain...**
 - a) people
 - b) fresh snow
 - c) ice and dust from when the planets formed
 - d) robots from ancient times
9. ☆ **In Poland, a Foucault pendulum showing the Earth's rotation can be found, for example, in...**
 - a) Kraków's Church of Sts. Peter and Paul
 - b) the port of Gdańsk
 - c) a factory in Łódź
 - d) Zakopane, on Gubałówka hill



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and look for sources of energy. The Sun's energy comes from the fusion of hydrogen into helium in its core. Fusion reactors such as ITER in France try to imitate the processes that occur in stars; ITER is being built jointly by countries from several continents. A tokamak holds hot plasma with a strong magnetic field. If fusion were mastered, humanity would have energy with almost no CO₂ emissions. The uranium in nuclear power plants was made in explosions of ancient stars, and gold, for which mines reach as deep as 4 km, was made in cosmic catastrophes involving stars. Helium, used in balloons and MRI, is rare on Earth and escapes into space. And geothermal heat comes partly from the decay of radioactive elements. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Sun's energy comes from...**
 - a) electricity
 - b) fusion of hydrogen into helium in its core
 - c) burning huge amounts of coal
 - d) fire
2. 👁 **Fusion reactors, such as ITER in France, try to imitate...**
 - a) windmills
 - b) volcanoes
 - c) lightning
 - d) the processes that occur in stars
3. 🗨 **The uranium in nuclear power plants was formed...**
 - a) in factories
 - b) in explosions of ancient stars
 - c) in the ocean
 - d) in mines
4. ⚡ **A tokamak is a device that holds hot plasma in place using...**
 - a) a strong magnetic field
 - b) water
 - c) ice
 - d) concrete walls
5. ⚡ **Gold mines reach as deep as 4 km underground, yet the gold itself was formed...**
 - a) in trees
 - b) in water
 - c) in cosmic catastrophes involving stars
 - d) from sand
6. 🧑 **The ITER project is being built jointly by...**
 - a) France alone
 - b) countries from several continents
 - c) private companies
 - d) the USA alone
7. ❤ **On Earth, the helium used in balloons and for cooling MRI magnets is...**
 - a) rare and escapes into space
 - b) inexhaustible
 - c) produced by plants
 - d) made from water
8. 🌿 **Geothermal energy from the Earth's interior comes partly from...**
 - a) the Sun
 - b) the decay of radioactive elements
 - c) the wind
 - d) the Moon
9. ☆ **If nuclear fusion could be mastered, humanity would have energy...**
 - a) only for the rich
 - b) from coal
 - c) only at night
 - d) with almost no CO₂ emissions

ASTRONOMY



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe, which, according to current measurements, is about 13.8 billion years old. The theory later called the Big Bang – the “primeval atom” hypothesis – was first proposed by Fr. Georges Lemaître. Today we know the expansion of the universe is accelerating, and its cause has been named dark energy. Dark matter and dark energy together make up about 95% of the universe. Our galaxy, the Milky Way, is a barred spiral, and in about 4.5 billion years it will collide with the Andromeda Galaxy. Its most numerous stars are red dwarfs. A star cluster is a group of stars that formed together from one cloud. Stars “pulse,” and asteroseismologists study the vibrations of the stars' interiors. A supernova is the explosion of a star at the end of its life. The first image of a black hole showed a dark shadow surrounded by a bright ring, and the black hole at the center of the Milky Way is called Sagittarius A*. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 Stars “pulsate” with sounds studied by helioseismologists and asteroseismologists. These are...
 - a) radio echoes
 - b) songs
 - c) vibrations of the stars’ interiors
 - d) wind
2. 👁 The first image of a black hole showed...
 - a) nothing at all
 - b) a dark shadow surrounded by a bright ring
 - c) a blue glowing sphere
 - d) a perfectly black square
3. 🗨 The black hole at the center of the Milky Way is called...
 - a) Sirius
 - b) Proxima
 - c) Andromeda
 - d) Sagittarius A*
4. 🕒 Roughly how old is the universe according to current measurements?
 - a) 100 billion years
 - b) 6,000 years
 - c) 4.5 billion years
 - d) 13.8 billion years
5. ✨ A star cluster is a group of stars that...
 - a) vanish during the day
 - b) are on Earth
 - c) glow green
 - d) formed together from one cloud
6. 🌌 In about 4.5 billion years, the Milky Way will collide with the...
 - a) Sombrero Galaxy
 - b) Whirlpool Galaxy
 - c) Triangulum Galaxy
 - d) Andromeda Galaxy
7. 💖 A supernova is...
 - a) the birth of a new planet
 - b) a new comet
 - c) the explosion of a star at the end of its life
 - d) a total solar eclipse
8. 🌿 The most numerous stars in the Galaxy are...
 - a) supernovae
 - b) blue giants
 - c) red dwarfs
 - d) white dwarfs
9. ⭐ Who first proposed the theory later called the Big Bang (the “primeval atom” hypothesis)?
 - a) Fr. Georges Lemaître
 - b) Stephen Hawking
 - c) Albert Einstein
 - d) Isaac Newton

1. c 2. b 3. d 4. d 5. d 6. d 7. c 8. c 9. a

A N S W E R S

MATHEMATICS



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and see how mathematics describes it. In 1922 Alexander Friedmann derived from Einstein's equations that the universe might be expanding. Einstein added a "cosmological constant" to his equations and later called it his biggest blunder. Kurt Gödel found a solution in which travel in time would be possible. The number 10^{80} is the estimated number of atoms in the observable universe. The distance to a nearby star is measured by the parallax method. The Hertzsprung-Russell diagram shows stars by brightness and temperature. Fourier's theorem breaks a telescope signal into frequency components. The arms of spiral galaxies resemble a logarithmic spiral. And a simple sum: light travels about 300,000 km/s, the Moon is about 300,000 km away, so light from the Moon takes about 1 second. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Fourier's theorem makes it possible to break down a telescope signal, like music, into its...**
 - a) Roman numerals
 - b) letters
 - c) paint colors
 - d) frequency components
2. 👁 **The Hertzsprung–Russell diagram arranges stars according to their...**
 - a) shape
 - b) sky color
 - c) age and name
 - d) brightness and temperature
3. 🗨 **The number 10^{80} is roughly the estimated number of...**
 - a) stars over Poland
 - b) atoms in the observable universe
 - c) grains of sand on a beach
 - d) people on Earth
4. ⚡ **Light travels at about 300,000 km/s, and the Moon is about 300,000 km from Earth (rounded). How many seconds does light take to travel from the Moon?**

5. 📏 **To measure the distance to a nearby star, astronomers measure its shift in the sky at different times of year. This is the method of...**
 - a) parallax
 - b) probing
 - c) weighing
 - d) echoes
6. 🧑 **In 1922, Alexander Friedmann derived from Einstein's equations that the universe might be...**
 - a) cube-shaped
 - b) flat as a table
 - c) expanding
 - d) eternal and never changing
7. 💖 **Einstein added a “cosmological constant” to his equations and later called it...**
 - a) a poem
 - b) his greatest success
 - c) a joke
 - d) his biggest blunder
8. 🌿 **Spiral galaxies have arms whose shape resembles a...**
 - a) logarithmic spiral
 - b) square
 - c) straight line
 - d) triangle
9. ⭐ **Kurt Gödel found a solution to Einstein's equations in which it would be possible to...**
 - a) travel underwater
 - b) travel in time
 - c) draw triangles
 - d) use negative numbers

OTHER (TECHNOLOGY)



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and see how technology brings it to everyone. The Webb telescope was named after a NASA administrator from the 1960s. The first full-color Webb image was unveiled in 2022 by US President Joe Biden. Color images from Webb are made by assigning colors to invisible wavelengths. The 2004 “Hubble Ultra Deep Field” revealed about 10,000 galaxies in a tiny patch of sky. Astronomers share telescope data publicly, with everyone. Sky-map apps let you identify stars with your phone, and to see galaxies through an amateur telescope it is best to travel far from city lights. Why is the night dark when there are so many stars? That is Olbers’ paradox. And the black hole Gargantua in the film “Interstellar” was calculated with the help of physicist Kip Thorne. Remember these details – I’ll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The film “Interstellar” (2014) showed the black hole Gargantua, whose image was calculated with the help of...
 - a) a painter
 - b) aliens
 - c) a child
 - d) physicist Kip Thorne
2. 👁 The “Hubble Ultra Deep Field” image (2004) showed, in a tiny patch of sky, about...
 - a) 10,000 galaxies
 - b) 1 planet
 - c) 10 stars
 - d) nothing
3. 🗨 The Webb telescope was named in honor of...
 - a) the inventor of the telephone
 - b) an astronaut who walked on the Moon
 - c) a soccer player
 - d) a NASA administrator from the 1960s
4. 🧮 Why is the night sky dark if there are so many stars? This is known as...
 - a) the Fermi paradox
 - b) the twin paradox
 - c) Olbers’ paradox
 - d) Zeno’s paradox
5. 🏠 To see distant galaxies through an amateur telescope, it is best to go...
 - a) far from city lights
 - b) into a tunnel
 - c) to a stadium at night
 - d) to a shopping mall
6. 🌐 Astronomers around the world share telescope data...
 - a) only for a fee
 - b) publicly, with everyone
 - c) never
 - d) only with governments
7. ❤ Sky map apps let you...
 - a) change the weather
 - b) switch off stars
 - c) call aliens
 - d) identify stars with your phone
8. 🍃 The color images from the Webb telescope are created by...
 - a) photographing a rainbow
 - b) using an ordinary camera
 - c) painting by hand
 - d) assigning colors to invisible wavelengths
9. ⭐ The first full-color image from Webb was presented in 2022 by...
 - a) the Pope
 - b) US President Joe Biden
 - c) the UN Secretary-General
 - d) the King of England

BIOLOGY

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe to understand the roots of life. Life on Earth is based on carbon, which formed in stars. Water is made of hydrogen, which formed just after the Big Bang, and every atom of iron in our blood was made in stellar explosions. In the Murchison meteorite scientists found amino acids, the building blocks of proteins. The panspermia hypothesis says the seeds of life may have come from space, e.g., on meteorites, though it remains unconfirmed. Astrobiology studies the possibility of life beyond Earth. It gets help from extremophiles in hot springs, which show where else life could exist, and from tardigrades, which survived the vacuum of space because they can dry out and “go to sleep.” Are we alone? So far we have no evidence of life beyond Earth. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Life on Earth is based on carbon, which was formed in...**
 - a) volcanoes
 - b) oceans
 - c) plants
 - d) stars
2. 👁 **Astrobiology is the science that studies...**
 - a) potted plants
 - b) animals in zoos
 - c) the possibility of life beyond Earth
 - d) birds in flight
3. 🗨 **The panspermia hypothesis says that the seeds of life may have come to Earth...**
 - a) from the ocean
 - b) from space, e.g., on meteorites
 - c) from a volcano
 - d) from the Moon on a rope
4. ⚗ **Water, essential for life, contains hydrogen, which was formed...**
 - a) in a factory
 - b) just after the Big Bang
 - c) in clouds
 - d) in plants
5. ✨ **Tardigrades have survived the vacuum of space because they can...**
 - a) breathe in a vacuum
 - b) dry out and “go to sleep”
 - c) glow
 - d) fly
6. 🦠 **In the Murchison meteorite, scientists found...**
 - a) living bacteria
 - b) the DNA of ancient dinosaurs
 - c) amino acids, the building blocks of proteins
 - d) fossilized tree leaves
7. 💜 **Every atom of iron in our blood was formed...**
 - a) in the liver
 - b) in water
 - c) in food
 - d) in stellar explosions
8. 🌿 **Extremophiles, organisms living in places such as hot springs, help us understand...**
 - a) where else life could exist
 - b) how to bake bread
 - c) how to build houses
 - d) how to play soccer
9. ☆ **Are we alone in the universe? So far, we...**
 - a) do not ask this question
 - b) have no evidence of life beyond Earth
 - c) know that we are not
 - d) have met aliens

CHEMISTRY

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe – to the birth of chemistry. In the first three minutes after the Big Bang, nuclei of hydrogen, helium and a little lithium formed – so the lithium in phone batteries partly formed just after the Big Bang. The first atoms, hydrogen and helium, formed about 380,000 years after the Big Bang. The most common element in the universe is hydrogen. This was described in a 1948 paper by Alpher and Gamow, who, as a joke so it would sound like “alpha, beta, gamma,” added Hans Bethe. Later the stars took over: the process of making elements in stars is nucleosynthesis. Star cores make elements up to iron, because heavier elements consume energy instead of releasing it. All the atoms in our bodies, except hydrogen, were made in stars. And in interstellar clouds water exists as ice on dust grains. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The first atoms (hydrogen and helium) formed about...
 - a) 380 years after the Big Bang
 - b) yesterday
 - c) 380 million years after the Big Bang
 - d) 380,000 years after the Big Bang
2. 👁 The most common element in the universe is...
 - a) carbon
 - b) iron
 - c) hydrogen
 - d) oxygen
3. 📖 The process by which elements are formed in stars is called...
 - a) photosynthesis
 - b) distillation
 - c) fermentation
 - d) nucleosynthesis
4. ⚖ In the cores of stars, elements form only up to iron, because...
 - a) iron is the last element in the alphabet
 - b) there is no room
 - c) heavier elements consume energy instead of releasing it
 - d) the stars get bored
5. ⚡ In the first three minutes after the Big Bang, the nuclei formed were those of...
 - a) uranium
 - b) iron
 - c) hydrogen, helium and a little lithium
 - d) gold
6. 🧑 The 1948 paper on the formation of elements was written by Alpher and Gamow, who as a joke, so the names would sound like “alpha, beta, gamma”, added...
 - a) Albert Einstein
 - b) Isaac Newton
 - c) Marie Curie
 - d) Hans Bethe
7. 💖 The lithium in phone batteries was partly formed...
 - a) in plants
 - b) in a mine
 - c) just after the Big Bang
 - d) in the ocean
8. 🌿 In interstellar clouds, water has been found in the form of...
 - a) rivers
 - b) ice on dust grains
 - c) seas
 - d) rain
9. ☆ All the atoms in our bodies, except hydrogen, were formed...
 - a) in stars
 - b) in a laboratory
 - c) in food
 - d) on Earth

PHYSICS



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe, into the world of physics. Right after the Big Bang, according to the cosmic inflation hypothesis, the universe expanded violently in a fraction of a second. Four forces were at work: gravity, electromagnetism, and the strong and weak interactions. Protons are built from the smallest particles, quarks. A “photon” is a particle of light, and light in a vacuum races at about 300,000 km/s. The universe is cooling because it is expanding – still expanding today – and the background radiation is now only about -270°C . When a massive star collapses under its own weight, a black hole forms. And gravitational waves from colliding black holes, turned into sound, are heard as a short “chirp.” Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Gravitational waves converted into sound sound like...**
 - a) thunder
 - b) a symphony
 - c) a short “chirp”
 - d) an hour of birdsong
2. 👁 **The smallest particles of matter that make up protons are...**
 - a) atoms
 - b) quarks
 - c) cells
 - d) molecules
3. 🗨 **The name “photon” refers to a particle of...**
 - a) light
 - b) water
 - c) heat
 - d) sound
4. ➡ **The universe is cooling because it is...**
 - a) evaporating like water
 - b) expanding
 - c) being cooled by someone
 - d) losing stars every day
5. ⚡ **Cosmic inflation is the hypothesis that just after the Big Bang the universe...**
 - a) expanded violently in a fraction of a second
 - b) froze solid in an instant
 - c) lost its monetary value
 - d) ceased to exist
6. 🌌 **The four fundamental forces of nature are gravity, electromagnetism and...**
 - a) magnetism and heat
 - b) the strong and weak interactions
 - c) wind and water
 - d) friction and elasticity
7. 💖 **The temperature of the background radiation is about...**
 - a) +270 °C
 - b) -270 °C
 - c) -27 °C
 - d) 0 °C
8. 🍃 **A black hole forms when...**
 - a) the Moon goes out
 - b) two comets collide head-on
 - c) a planet explodes
 - d) a massive star collapses under its own weight
9. ⭐ **The speed of light in a vacuum is about...**
 - a) 30 km/s
 - b) 300 km/h
 - c) 3,000 km/s
 - d) 300,000 km/s

ENVIRONMENT

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and explore the environment where stars were born. The first stars lived in a universe made almost entirely of hydrogen and helium. Stars are born in nebulae, and a nebula is a cloud of gas and dust in space. Our Solar System formed from a cloud of gas and dust that collapsed – perhaps triggered by a nearby supernova, and from it came the Sun. Other stars formed from the same cloud too – the Sun's “siblings.” Rocky planets formed close in, because there it was too hot for ice and gases. The asteroid belt is leftover material that never formed a planet, and the Oort Cloud is a distant cloud of icy bodies. A star's habitable zone is the region where a planet can have liquid water. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The “environment” of the first stars was a universe made almost entirely of...
 - a) water
 - b) hydrogen and helium
 - c) iron
 - d) carbon dust
2. 👁 A nebula is...
 - a) a cloud of gas and dust in space
 - b) a type of spiral galaxy
 - c) a rain cloud
 - d) fog over a lake
3. 🗨 A star’s habitable zone (sometimes called its ecosphere) is the region where...
 - a) there are no planets
 - b) there are the most comets
 - c) a planet can have liquid water
 - d) it is darkest
4. ⚙ The Solar System formed from...
 - a) a cloud of gas and dust that collapsed
 - b) the Moon
 - c) a second Sun
 - d) an exploding Earth
5. ⚡ Rocky planets formed close to the Sun because there...
 - a) it was cold
 - b) there was no dust
 - c) it was too hot for ice and gases
 - d) there was an ocean
6. 🧑 The explosion of a nearby supernova may have triggered the collapse of the cloud that formed...
 - a) the Sun
 - b) Andromeda
 - c) the Milky Way
 - d) the Moon
7. ❤ The same cloud that formed the Sun also produced...
 - a) dinosaurs
 - b) other stars – the Sun’s “siblings”
 - c) oceans
 - d) forests
8. 🌿 The asteroid belt between Mars and Jupiter is...
 - a) leftover material that never formed a planet
 - b) a destroyed planet of humans
 - c) the Oort Cloud
 - d) the Kuiper Belt
9. ☆ The Oort Cloud is...
 - a) a distant cloud of icy bodies
 - b) a galaxy
 - c) a planet
 - d) a constellation

1.b 2.a 3.c 4.a 5.c 6.a 7.b 8.a 9.a

A N S W E R S

HEALTH

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and see what space has to do with health. The calcium in our bones was made in stars, and calcium, iron and iodine all have a cosmic origin – from stars. We cannot hear stars explode because sound does not travel in a vacuum. The Sun heals and harms: vitamin D forms in the skin under sunlight, but UV can hurt the skin, so in summer we use sunscreen. Looking at the Sun through a telescope without a filter can permanently damage your eyesight. Earth's atmosphere and magnetic field protect us from cosmic radiation. Our daily rhythm depends on the Earth's rotation and light. And astronauts returning from space must rebuild their muscle strength. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Why can't we hear stars exploding? Because...**
 - a) our ears are weak
 - b) sound does not travel in a vacuum
 - c) they are too quiet
 - d) we are asleep
2. 👁 **Looking at the Sun through a telescope without a filter can...**
 - a) do nothing
 - b) permanently damage your eyesight
 - c) improve your eyesight
 - d) cure a cold
3. 🗨 **Vitamin D is produced in the skin under the influence of...**
 - a) darkness
 - b) wind
 - c) rain
 - d) sunlight
4. ⚡ **The Earth's atmosphere and magnetic field protect us from...**
 - a) sound
 - b) cosmic radiation
 - c) gravity
 - d) rain
5. 🏃 **After returning from space, astronauts have to...**
 - a) run a marathon right away
 - b) sleep for a week
 - c) eat only fish
 - d) rebuild their muscle strength
6. 🧑 **UV radiation from space can harm the skin, so in summer we...**
 - a) use sunscreen
 - b) stop eating
 - c) avoid water
 - d) sleep during the day
7. 💖 **The body's daily rhythm depends on...**
 - a) the rain
 - b) the wind
 - c) the Earth's rotation and light
 - d) the phases of Mars
8. 🌿 **The calcium in our bones was formed in...**
 - a) water
 - b) stars
 - c) rocks
 - d) milk
9. ☆ **Calcium, iron and iodine, needed for good health, have an origin that is...**
 - a) plant-based only
 - b) from water
 - c) cosmic – from stars
 - d) from the store

PHYSICAL CULTURE AND SPORT

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and join a sports practice in space. On the ISS, astronauts run on a treadmill strapped in with elastic harnesses. A ball kicked in weightlessness flies straight until it hits something. In weightlessness the spine straightens out, so the body grows a few centimeters longer. On the Moon gravity is 6 times weaker, so a long jump would be longer, and someone weighing 60 kg on Earth would see about 10 kg on a scale. In 1971 Alan Shepard hit golf balls there. To become an astronaut you need good fitness and mental resilience, and a crew must cooperate for months. Sławosz Uznański, the second Pole in space, flew to the ISS in 2025. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Astronauts on the ISS exercise on a treadmill, strapped in with...**
 - a) ropes to the wall
 - b) magnets
 - c) elastic harnesses
 - d) duct tape
2. 👁️ **A ball kicked in weightlessness...**
 - a) falls to the floor
 - b) stops
 - c) returns to your foot
 - d) flies straight until it hits something
3. 🗨️ **The longest long jump on the Moon would be longer than on Earth because...**
 - a) there is wind there
 - b) it is dark
 - c) gravity is 6 times weaker
 - d) it is cold there
4. ⚖️ **If someone weighed 60 kg on Earth, a scale on the Moon would show about...**
 - a) 60 kg
 - b) 0 kg
 - c) 10 kg
 - d) 360 kg
5. 🏈 **In 1971, astronaut Alan Shepard, while on the Moon,...**
 - a) played soccer
 - b) hit golf balls
 - c) ran a marathon
 - d) went swimming
6. 🧑‍🤝‍🧑 **Teamwork is important in space because the crew...**
 - a) plays soccer every day
 - b) must cooperate for months
 - c) competes for food
 - d) does not talk to each other
7. 💖 **To become an astronaut, you need...**
 - a) connections
 - b) only a diploma
 - c) good fitness and mental resilience
 - d) a lot of money
8. 🍀 **Sławosz Uznański, the second Pole in space, flew to the ISS in...**
 - a) 1978
 - b) 2025
 - c) 2005
 - d) 2015
9. ☆ **In weightlessness the body gets a few centimeters taller because...**
 - a) the bones grow
 - b) the muscles swell
 - c) the spine straightens out
 - d) clothes shrink

MEDICINE

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and see how its secrets help medicine. Liquid helium, made mostly just after the Big Bang, cools the superconducting magnets in MRI scanners. Radioactive isotopes for nuclear medicine are made in reactors and accelerators; iodine-131 is used to treat the thyroid. Radiotherapy destroys cancer cells using radiation, and Marie Skłodowska-Curie founded institutes where radiation was used for treating cancer. Space teaches doctors too. Research on the ISS helps treat osteoporosis (bone weakening). An astronaut's brain shifts slightly inside the skull. Scott Kelly spent a year in space while his twin brother stayed on Earth – the comparison showed changes in his genes and health. And to fly to Mars, we must solve the problem of cosmic radiation. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **An astronaut's brain in space...**
 - a) doubles in size
 - b) shrinks by half
 - c) stops working
 - d) shifts slightly inside the skull
2. 👁️ **Radioactive isotopes used in nuclear medicine are produced...**
 - a) in forests
 - b) in the ocean
 - c) in reactors and accelerators
 - d) in refrigerators
3. 🗨️ **Research on the ISS helps treat conditions on Earth such as...**
 - a) the common cold
 - b) only toothache
 - c) osteoporosis (bone weakening)
 - d) a broken heart
4. ⚡ **MRI scanners need liquid helium, which...**
 - a) supplies electricity
 - b) colors the image
 - c) cools the superconducting magnets
 - d) puts the patient to sleep
5. ⚡ **Radiotherapy destroys cancer cells using...**
 - a) warm water
 - b) sound
 - c) radiation
 - d) cold
6. 👩 **Marie Skłodowska-Curie founded institutes where radiation was used for...**
 - a) heating
 - b) cooking
 - c) treating cancer
 - d) lighting streets
7. ❤️ **Astronaut Scott Kelly spent a year in space while his twin brother stayed on Earth. The comparison showed...**
 - a) that he had not changed
 - b) that he forgot how to talk
 - c) changes in his genes and health
 - d) that he grew 1 m
8. 🌿 **The isotope iodine-131 is used to treat the...**
 - a) teeth
 - b) thyroid
 - c) eyes
 - d) bone fractures
9. ⭐ **To fly to Mars, we need to solve the problem of...**
 - a) smells
 - b) rain
 - c) boredom
 - d) cosmic radiation

PHARMACY



1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe and peek into a cosmic pharmacy. Elements needed for medicines, such as lithium and platinum, were formed in cosmic processes. Glycine has also been found in space – it is one of the amino acids. Radiopharmaceuticals are drugs that contain radioactive isotopes. The half-life tells how long it takes for half of the atoms to decay, so isotopes must be transported quickly, because they decay. The Maria research reactor in Świerk produces isotopes for hospitals around the world. On the ISS, medicines lose potency faster, probably because of cosmic radiation. But protein crystals grow larger and more regular in weightlessness, which helps design drugs. And for long missions, scientists hope in the future to print medicines on board. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Medicines on the ISS lose their effectiveness faster, probably because of...
 - a) silence
 - b) darkness
 - c) music
 - d) cosmic radiation
2. 👁 Protein crystals grown in weightlessness are...
 - a) not formed at all
 - b) larger and more regular
 - c) colorful
 - d) smaller
3. 📖 Radiopharmaceuticals are drugs that contain...
 - a) herbs
 - b) radioactive isotopes
 - c) sugar
 - d) vitamins
4. ⚖ The half-life of an isotope tells us how long it takes for...
 - a) half of the atoms to decay
 - b) a drug to dissolve
 - c) a pill to change color
 - d) a patient to fall asleep
5. ⚡ To get isotopes to a hospital in time, they must be...
 - a) sent by regular mail
 - b) transported quickly, because they decay
 - c) stored for years
 - d) frozen for a year
6. 🧑🔬 The Maria research reactor in Świerk, Poland, produces isotopes...
 - a) for refrigerators
 - b) for watches
 - c) for hospitals around the world
 - d) for cars
7. 💜 Organic molecules found in space, such as glycine, are...
 - a) antibiotics
 - b) vitamins
 - c) painkillers
 - d) amino acids
8. 🌿 Elements needed for medicines, such as lithium or platinum, were formed...
 - a) in cosmic processes
 - b) in plants
 - c) only in laboratories
 - d) in pharmacies
9. ⭐ For long space missions, scientists would like in the future to...
 - a) print medicines on board
 - b) ask aliens for help
 - c) bring whole pharmacies
 - d) take no medicines

1.d 2.b 3.b 4.a 5.b 6.c 7.d 8.a 9.a

A N S W E R S

AGRICULTURE

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and visit a cosmic field. The elements plants need – nitrogen, phosphorus, potassium – were made in stars; phosphorus, important in fertilizers, formed in explosions of massive stars. Farming on Earth depends on the Sun, whose energy comes from nuclear fusion. In space, crops matter because on long missions the crew needs fresh food. In 2015 astronauts on the ISS grew and ate lettuce. Plants there grow under LED lamps in red and blue, and in weightlessness their roots grow in different directions. In 2019 China's Chang'e 4 probe on the Moon sprouted a cotton seed. And Martian soil, regolith, is unusable without preparation because it contains harmful salts. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In 2015, astronauts on the ISS grew and ate...
 - a) lettuce
 - b) wheat
 - c) corn
 - d) potatoes
2. 👁 Plants in space grow under LED lamps in the colors...
 - a) red and blue
 - b) green
 - c) yellow
 - d) white only
3. 🗨 All the elements plants need, such as nitrogen, phosphorus and potassium, were formed...
 - a) in stars
 - b) in rain
 - c) in fertilizers
 - d) in the soil
4. ⚖ In weightlessness, plant roots grow...
 - a) not at all
 - b) always downward
 - c) in different directions
 - d) only upward
5. ⚡ Martian soil (regolith) is not suitable for growing crops without preparation because...
 - a) it is made of plastic
 - b) it contains harmful salts
 - c) it is too wet
 - d) it is fertile
6. 🌕 In 2019, China's Chang'e 4 probe on the Moon...
 - a) planted trees
 - b) harvested potatoes
 - c) raised cows
 - d) sprouted a cotton seed
7. ❤ Farming on Earth depends on the Sun, whose energy comes from...
 - a) coal
 - b) water
 - c) nuclear fusion
 - d) wind
8. 🌿 Phosphorus, important in fertilizers, was formed in...
 - a) the soil
 - b) plants
 - c) seawater
 - d) explosions of massive stars
9. ⭐ Growing crops in space is important because on long missions...
 - a) plants look nice
 - b) the crew needs fresh food
 - c) there is nothing else to do
 - d) plants produce fuel

VETERINARY SCIENCE

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and meet the animal travelers of space. Animals were studied before humans to check that spaceflight was safe. Dogs, monkeys, tortoises, and also spiders and fish flew into space. In 1968, steppe tortoises aboard Zond 5 became the first animals to circle the Moon and come back. Spiders on the ISS wove webs less symmetrical than on Earth, and fish swam in circles, losing their orientation. Mice in special modules help study bone and muscle loss. The dog Laika never returned; years later her handlers regretted that she could not be saved. Today veterinarians work with space agencies to look after the welfare of research animals. And on Earth, the magnetic field helps birds with navigation during migration. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦋 **Animals sent into space have included dogs, monkeys, tortoises and also...**
 - a) elephants
 - b) horses
 - c) spiders and fish
 - d) giraffes
2. 🕷 **Spiders on the ISS spun webs in weightlessness that were...**
 - a) less symmetrical than on Earth
 - b) identical
 - c) not spun at all
 - d) square
3. 🐢 **Steppe tortoises on the Zond 5 probe (1968) were the first animals to...**
 - a) circle the Moon and come back
 - b) fly to the Sun
 - c) stay on the Moon
 - d) land on Mars
4. 🚀 **Animals were tested in space before humans in order to...**
 - a) make them astronauts
 - b) scare aliens
 - c) sell them
 - d) check that spaceflight was safe
5. 🐟 **Fish in weightlessness swim...**
 - a) not at all
 - b) normally
 - c) always upside down
 - d) in circles, losing their orientation
6. 🐾 **Veterinarians work with space agencies to...**
 - a) breed farm animals on Mars
 - b) sell animals to zoos
 - c) feed astronauts meat
 - d) look after the welfare of research animals
7. 🐭 **Mice in special modules on the ISS help scientists study...**
 - a) bone and muscle loss
 - b) the taste of cheese
 - c) running speed
 - d) fur color
8. 🌿 **The Earth's magnetic field helps animals such as birds with...**
 - a) eating
 - b) sleeping
 - c) singing
 - d) navigation during migration
9. 🌟 **The dog Laika never came back from space. Years later, the people who cared for her...**
 - a) hid the story
 - b) forgot about her
 - c) regretted that she could not be saved
 - d) were proud and had no regrets



ANIMAL SCIENCE AND FISHERIES

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago

Let's travel back to the beginning of the universe – the iron in the hemoglobin of fish and mammals was made in stars. Today animals and insects keep us company in orbit. Medaka fish on the ISS help study bone changes in weightlessness. Hawaiian bobtail squid were studied because they glow thanks to bacteria. Fruit flies show how weightlessness affects genes and the heart. Bees aboard the space shuttle in 1984 learned after a few days to build honeycombs in weightlessness. In the future, protein on space stations may come from insects and algae. Teams from many countries raise organisms together, because space on the station is valuable and shared. Animals must have food, water and good welfare, and decisions about flying animals weigh animal welfare and the value of the research. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 **Medaka fish on the ISS help scientists study...**
 - a) colors
 - b) singing
 - c) bone changes in weightlessness
 - d) the weather
2. 👁 **Hawaiian bobtail squid were studied on the ISS because they...**
 - a) are very large
 - b) can talk
 - c) eat metal
 - d) glow thanks to bacteria
3. 📖 **Insects such as fruit flies help scientists in space study...**
 - a) how weightlessness affects genes and the heart
 - b) how to teach insects to fly faster
 - c) the weather
 - d) the taste of fresh fruit
4. 🐷 **Protein for food on future space stations could come from...**
 - a) insects and algae
 - b) cows
 - c) pigs
 - d) caged chickens
5. 🐝 **Bees on the space shuttle (1984), after a few days,...**
 - a) escaped
 - b) turned into butterflies overnight
 - c) stopped flying forever
 - d) learned to build honeycombs in weightlessness
6. 🌐 **Teams of scientists from many countries raise organisms on the ISS together because...**
 - a) it is cheap
 - b) space on the station is valuable and shared
 - c) nobody else wants to do it
 - d) it is forbidden
7. ❤ **Raising animals in space requires that they be provided with...**
 - a) no room to move
 - b) lots of noise
 - c) food, water and good welfare
 - d) constant darkness
8. 🌱 **The iron in the hemoglobin of fish and mammals was formed...**
 - a) in ice
 - b) in plants
 - c) in stars
 - d) in water
9. ☆ **Should animals be sent into space? Today, the decision takes into account...**
 - a) only the cost of the flight
 - b) drawing lots
 - c) current fashion
 - d) animal welfare and the value of the research

OTHER (NATURE)

1. Beginning of the Universe

13.8 bn years ago – 4.5 bn years ago



Let's travel back to the beginning of the universe and gaze at the night sky. A Greek myth says the goddess Hera spilled milk across the sky – hence the “Milky Way.” It is best to observe it on a moonless night far from cities. When we look at the stars we look into the past, because their light took a long time to reach us. The Andromeda Galaxy, the most distant object visible to the naked eye, lies about 2.5 million light-years away. Scientists from many countries determined the age of the universe by comparing data from satellites and telescopes. Our Sun is a star of average size, and its vibrations, ringing like a bell, are studied by helioseismologists. The solar wind causes the aurora – it is its visible effect. And the cosmic scale teaches us that Earth is a small and precious planet. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Sun “rings” like a bell – its vibrations are studied by...**
 - a) helioseismologists
 - b) meteorologists
 - c) geologists
 - d) musicians
2. 👁 **The aurora is a visible effect of...**
 - a) the solar wind
 - b) the Moon’s reflection
 - c) a meteor shower
 - d) glowing ice
3. 🗨 **The name “Milky Way” comes from a myth in which the goddess Hera...**
 - a) lit a fire
 - b) planted a tree
 - c) lost a ring
 - d) spilled milk across the sky
4. ⚖ **The Sun is a star that is...**
 - a) the largest in the universe
 - b) the smallest
 - c) already burned out
 - d) of average size
5. ✨ **To see the Milky Way, it is best to...**
 - a) look out a window in the city
 - b) observe on a moonless night far from cities
 - c) look during the day
 - d) use a flashlight
6. 🌍 **Scientists from many countries determined the age of the universe together by comparing data from...**
 - a) legends
 - b) dreams
 - c) books
 - d) satellites and telescopes
7. 💖 **Looking at the stars, we look into the past because...**
 - a) poets say so
 - b) the stars are old
 - c) clocks stop
 - d) their light took a long time to reach us
8. 🌿 **The Andromeda Galaxy is the most distant object visible to the naked eye – it lies about...**
 - a) 25 km away
 - b) 2.5 million light-years away
 - c) 2.5 billion km away
 - d) 250 light-years away
9. ⭐ **The cosmic scale teaches us that the Earth is...**
 - a) a small and precious planet
 - b) the largest planet
 - c) the center of the universe
 - d) indestructible

1.a.2.a.3.b.4.d.5.b.6.d.7.d.8.b.9.a

ANSWERS